

Thermodynamics

Lecture 1A:

- Course plan and rules
- Motivation

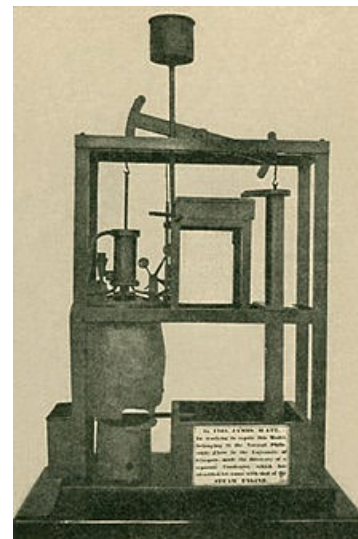
Prof. Claudio Hail (he/him)

Course theme

- ‘Thermodynamics can be defined as the science of energy’
AY Cengel
- From Greek: *therme* (heat) & *dynamis* (power)
- Includes energy, energy transformations, power conversion, refrigeration, relationships of material properties and more



Image source: shutterstock, wikipedia.org
C. Hail – ME40 – Fa26



Steam engine

Teaching team



Head GSI:
Cassidy McCormick



GSI:
Jessica Wu



Reader:
Ninad Atale



Reader:
Takenori Tsuchida

Learning objectives (I)

What you will be able to do after this course:

- I. *Knowledge*: Write the first and second laws of thermodynamics.
- II. *Comprehension*: Describe the first and second laws in the student's own language.
- III. *Application*: Solve simple single-answer problems using the first law.
- IV. *Analysis*: Solve problems requiring both the first and second laws.
- V. *Synthesis*: Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

Learning objectives (II)

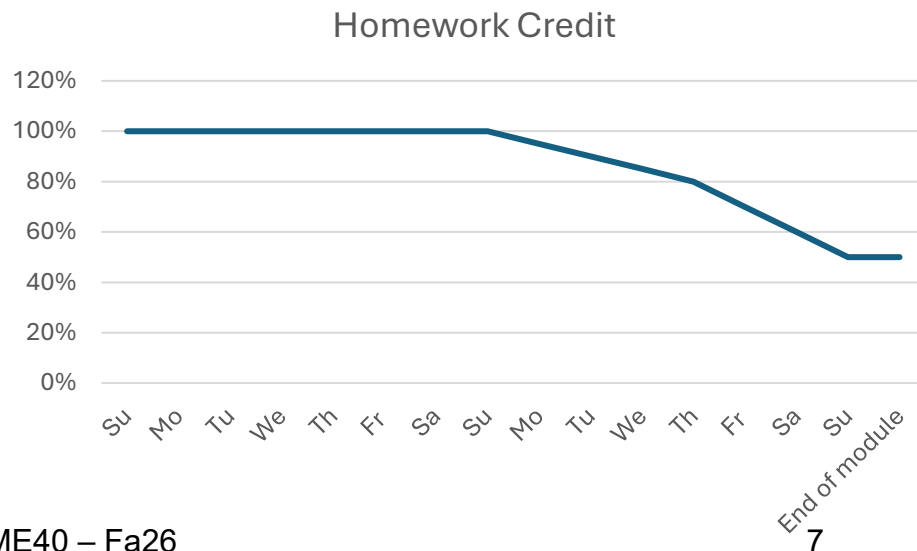
- VI. *Evaluation*: Determine and describe second law fallacies in proposed power cycles.
- VII. *Evaluation*: Judge when classical thermodynamics is not the appropriate analysis tool.
- VIII. *Evaluation*: Find and correct errors in the student's own solutions and in those of others.
- IX. *Analysis and Evaluation*: Search appropriate databases and the literature to find required thermodynamic data, and if the data are not available the student can select appropriate procedures and predict the values of the data.
- X. *Knowledge*: Briefly outline the history of the field of thermodynamics.

Homework (I)

- Homework assignments will be posted in [PrairieLearn](#) (online learning and assessment platform).
- Please enroll in our course here: [PrairieLearn](#)
- Submit answers and get immediate feedback.
- Regular problems: Unlimited trials are allowed without penalty for wrong answers.
- High-stakes problem: Only single trial is allowed.
- For full credit: upload scanned written notes to PrairieLearn.

Homework (II)

- Homework assignments will be posted each Sunday and are due the Sunday of the following week.
- Late homework submission is penalized as shown in the graph below.
- The lowest scoring homework will be dropped from the final grade calculation.



Some remarks on the Homework

HW1: Homework 1

Total points: 0/11

0%

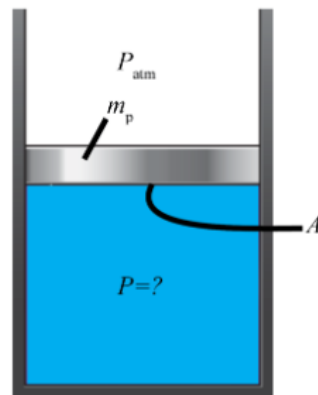
 Assessment is **open** and you can answer questions.
Available credit: 100% (Staff override) ●

Comments about this assessment:
Download this [answer booklet](#) and read the instructions.
You must upload a digital PDF copy of your work, using this booklet or another means, in the section below labeled **Personal Notes** and the filename should be [firstname].[Lastname]HW[number]Fa25.pdf, i.e., sadiCarnotHW7Fa25.pdf.
Failure to upload your work may result in a full loss of points for the assessment.

Question	Value	Variant history	Awarded points
#847. Resistance Heater Water	1		— /1
#108. Pressure Gauge Liquid	1		— /1
#942. Pressure Gauge	1		— /1
#977. Isothermal, Isobaric, Isochoric Processes	1		— /1
#777. Direction of Heat Transfer	1		— /1
#936. Temperature Conversion	1		— /1
#258. Piston with Weight	1		— /1
#826. JupyterLab and pyCalor	1		— /1
#181. Extensive vs. Intensive Properties	1		— /1
#172. State Postulate	1		— /1
#521. Plastic Tank Weight	1		— /1

#258. Piston with Weight

The piston of a vertical piston-cylinder device containing a gas has a mass of $m_p = 70 \text{ kg}$ and a cross-sectional area of $A = 0.04 \text{ m}^2$. The local atmospheric pressure is $P_{\text{atm}} = 110 \text{ bar}$, and the gravitational acceleration is 9.81 m/s^2 . Determine the pressure inside the cylinder in units of bar (1 bar = 100,000 Pa).



Learning management system: bCourses

- We will communicate to you through announcements.
- **Please communicate to us through email with the subject line beginning: “ME40:”**
 - **GSIs are first point of contact**
- bCourses discussions are not frequently monitored.

▼ Module 1

📄 Syllabus

📄 1. Introduction and basic concepts (8/26, 8/28)

▼ Module 2

▼ Module 3

▼ Project and Final Exam

📊 View Course Stream

📅 View Course Calendar

🔔 View Course Notifications

To Do

🔔 [Welcome to Thermodynamics ME40!](#)
Thermodynamics (Fall 2026)
Aug 25 at 1:34pm

bCourses: Modules and Pages

- Clicking on a page in the module

▼ Module 1
 Syllabus
 1. Introduction and basic concepts
 2. Energy, Energy Transfer, and General Energy Analysis
 3. Properties of Pure Substances
 4. Energy Analysis of Closed Systems
 5. Mass and Energy Analysis of Control Volumes

To do before lecture:



To do by Sunday
following week:

To read for this week:

1. Introduction and basic
concepts

Mandatory pre-work

Mandatory pre-work will be posted here.

Lecture slides

The lecture slides will be posted here.

Homework

You can find your first assignment here: <http://us.prairielearn.com/>

Mandatory Reading

Chapter 1: Introduction and Basic Concepts from Thermodynamics: An Engineering Approach by Cengel and Boles.

bCourses: Modules and Pages

- The course has three “modules” with each four to six “pages”, exams are at the end of each module.

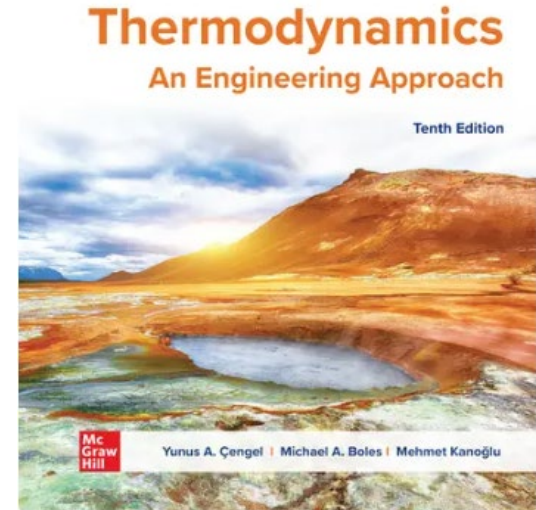
▼ Module 1
 Syllabus
 1. Introduction and basic concepts
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 3. Properties of Pure Substances
 4. Energy Analysis of Closed Systems
 5. Mass and Energy Analysis of Control Volumes

▼ Module 2
 6. Second Law of Thermodynamics
 7. Entropy
 8. Exergy
 9. Gas Power Cycles

▼ Module 3
 10. Vapor and Combined Power Cycles
 11. Refrigeration Cycles
 12. Thermodynamics Property Relations
 13. Gas Mixtures
 14. Gas-Vapor Mixtures and Air Conditioning
 15. Chemical Reactions

Textbook

- Thermodynamics: an engineering approach / Y. A. Cengel.
- Available online [here](#) through Berkeley Library
- Other recommended books:
 - *Fundamentals of engineering thermodynamics*, Moran & Shapiro



Lecture recording

- All lectures will be recorded and made available on bCourses.
- Group or plenary discussions will be removed/trimmed before uploading to allow for more open discussion.
- At times audio/visual equipment may not work and we may be unable to upload the content.

Midterms, final and project

- Midterm 1: 10/5/26
- Midterm 2: 11/9/26
- Final Exam: 12/17/26
- Thermodynamics project

Midterm 1	40 pts
Midterm 2	40 pts
Final Exam	70 pts
Project	20 pts
Homework	30 pts
Total	200 pts

Exam recovery:

$$M_{1,final} = \max(M_{1,original} \cdot 0.6 + Final \cdot 0.4, M_{1,original})$$

Discussion Sessions

- 101: Tuesday, 5:00 pm – 6:00 pm in Stanley 106, Jessica Wu
- 102: Thursday, 1:00 pm – 2:00 pm in Etcheverry 3111, Cassidy McCormick
- 103: Friday, 12:00 pm – 1:00 pm in Etcheverry 3107, Cassidy McCormick

Office hours

- Monday, 10 am – 11 am, Hesse Hall “Fishbowl”, GSI Jessica Wu
- Wednesday 1 pm – 2 pm, Etcheverry 6173, Prof. Claudio Hail
- Thursday, 2:30 pm – 3:30 pm, Hesse Hall “Fishbowl”, GSI Cassidy McCormick

Use of Generative AI

- Use of Gen AI is discouraged in this course.
- Assignments are designed to foster your critical thinking, problem solving, and subject matter understanding.
- Submitting AI-generated material as your work violates course academic integrity policy and may result in disciplinary action.
- If you must, use it as a study helper:
 - Practice problem generation
 - Socratic tutor: *I'm working on a problem involving Act as a Socratic tutor. Do NOT give me the solution or final answer. Ask me one guiding question at a time to help me figure out the next step.*
 - Generate a summary of course material

CalorAI: Thermodynamics AI Study Helper

- We are currently developing an AI study helper for this course.
- For interested students: Testing of the AI Agent is planned for a 2-week period later in the semester.
- For more information:
 - <https://engineering.berkeley.edu/news/2026/01/uc-berkeley-professors-explore-new-ways-to-integrate-ai-in-engineering-courses/>

Course Schedule (I)

Week	Su	Mo	Tue	Wed	Thu	Fri	Sat
1		8/24 No Class					
2	HW1 posted	8/31					
3	HW2 posted	9/7 Holiday					
4	HW1 and HW2 due, HW3 posted	9/14					

Course Schedule (II)

Week	Su	Mo	Tue	Wed	Thu	Fri	Sat
5	HW3 due HW4 posted	9/21				HW4 due HW5 posted	
6	HW4 due HW5 posted	9/28					
7	HW5 due	Midterm II 10/5					
8	HW6 posted	10/12					

Course Schedule (III)

Week	Su	Mo	Tue	Wed	Thu	Fri	Sat
9	HW6 due HW7 posted	10/19					
10	HW7 due HW8 posted	10/26					
11	HW8 due HW9 posted	11/2					
12	HW9 due	Midterm II 11/9		Holiday			

Course Schedule (IV)

Week	Su	Mo	Tue	Wed	Thu	Fri	Sat
13	HW10 posted	11/16					
14	HW10 due HW11 posted	11/23		No Class	Holiday	Holiday	
15	HW11 due HW12 posted	11/30					

Course Schedule (V)

Week	Su	Mo	Tue	Wed	Thu	Fri	Sat
16	HW12 due Project due	12/7	RRR	RRR	RRR	RRR	
17		12/14			Final Exam		

Course Feedback

- Include more worked through examples in lectures.
- Homework does not prepare you for midterm/final.
- Large workload with homework, project, and exams.
- Extremely difficult exams, long and tedious homework.
- Make classes more engaging.

How to do well in this class

- Complete textbook readings before the class.
- Come to class and discussions.
- Don't be shy, ask questions come to office hours.

Thermodynamics

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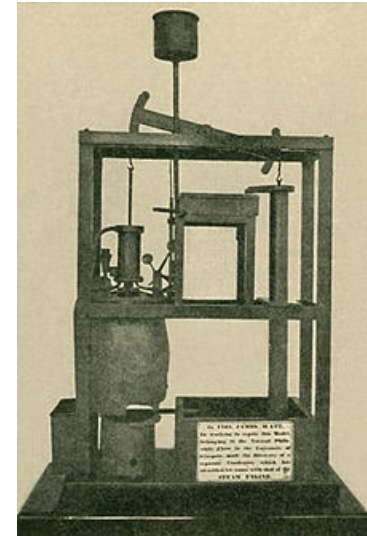
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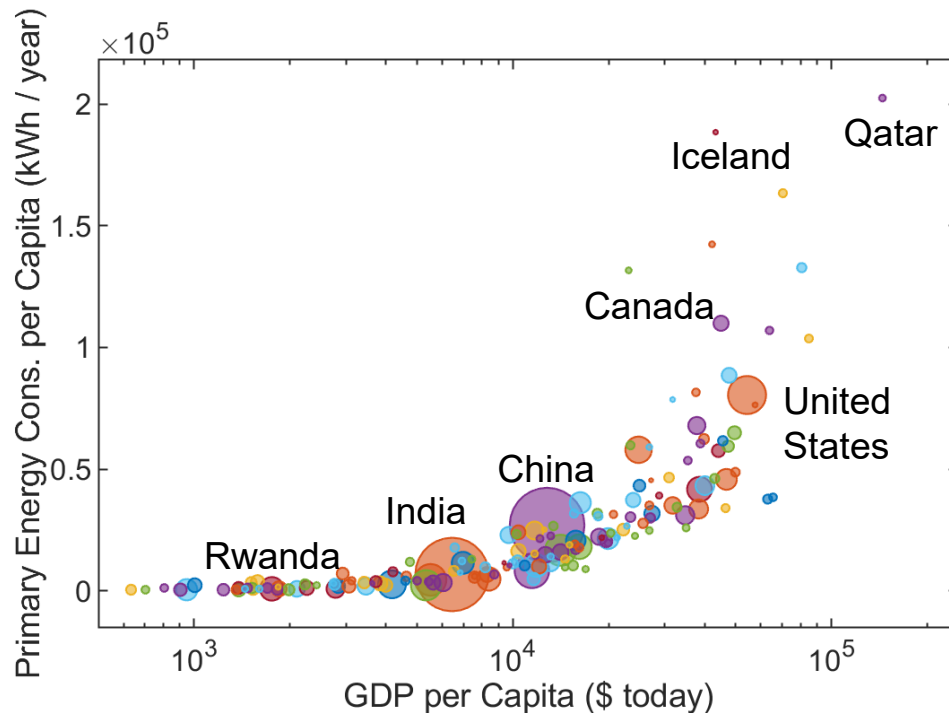


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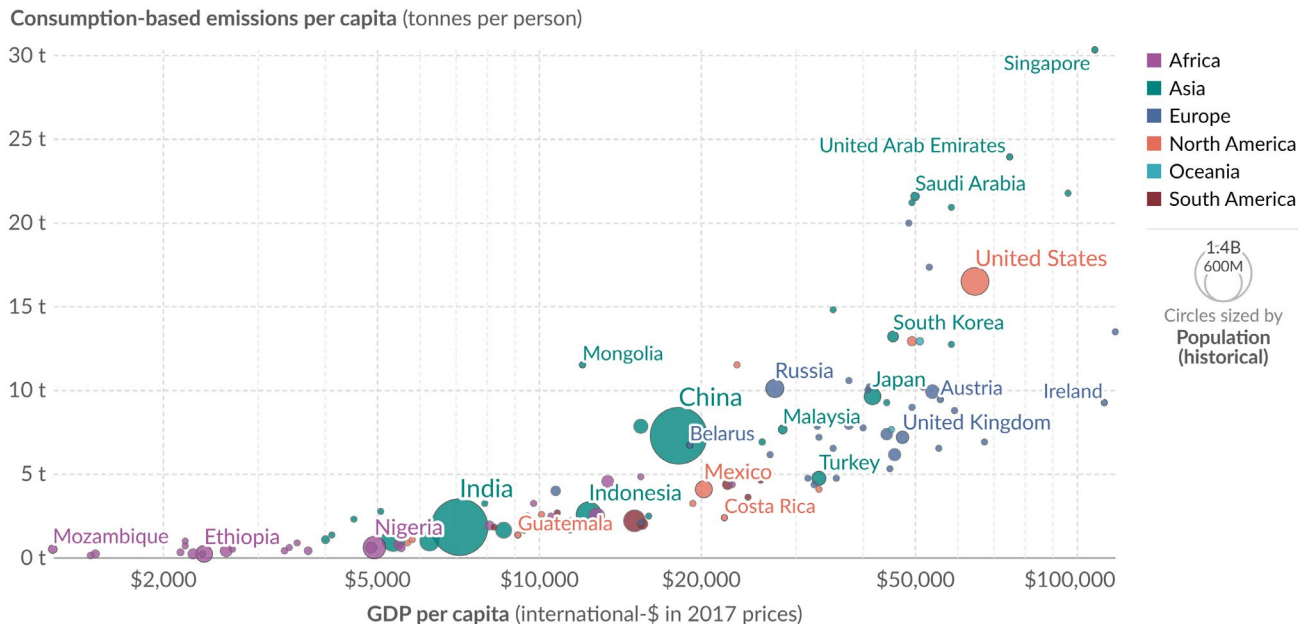


Steam engine

Global energy demand



Energy and CO₂ emissions



Data source: Global Carbon Budget (2024); Population based on various sources (2024); World Bank (2023)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Thermodynamics: relevance



Airplanes



Power plants



Cars

Thermodynamics: relevance



Food processing



Refrigeration



Heating

Thermodynamics challenges in the future

- Electrification
- Decarbonization
- Artificial Intelligence
- Emerging technologies:
 - Atmospheric water harvesting
 - Passive radiative cooling
 - ...

Next steps

- Login and check out bCourses and check out.
- Login and enroll in PrairieLearn ME40.
- Complete reading for week 1 in Cengel's book.

